A constant variance bound removes the distant count tail

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The incoming counts need not be assigned variances growing with their distance from the own count. Their variances have a common bound. Using that bound after the first one or two counts produces a factorial tail, and extends most of the direct comparison to $ns \le .075$. The only phase interval left out of this note is $-.15 < u < 0$.

Theorem 1. *Let $n \ge 42$, $2/5 \le p \le 53/100$, $q = 1-p$, and $Y = B + U$, where $\mathbb{P}(B=1) = p$ and $\mathbb{E}(U \mid B) = 0$. The two conditional laws satisfy*

$$|U_j| \le 3/5, \quad U_0 \le a_0, \quad U_1 \ge -a_1, \qquad a_j \ge 0, \quad a_0 + a_1 \le 9/125.$$

Put $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (Y-p)/\sqrt{v+s}$, $\rho = \mathbb{E}|X|^3$, $\rho_B = (1-2v)/\sqrt{v}$, and $U_* = 3/40$. Suppose $0 < \lambda \le U_*$, $|H| \le 46/25$, and $C \ge 2/5$. Choose $k \in \mathbb{Z}$ with $np + H\sqrt{v+s} = k + u$, where

$$u \in [-34/125, -3/20] \ \cup \ [0, 91/125].$$

Let $\delta = k - np$, let F_B be the $\mathrm{Bin}(n,p)$ CDF, and let S_n be the sum of n independent copies of X. Then

$$\begin{aligned}\Delta_C = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B < -\lambda/100.\end{aligned}$$

The theorem assumes the conditional representation and support bounds; it does not acquire them. In particular it is not a full Berry–Esseen theorem, nor does it settle the omitted negative phase. The old $ns \le .03$ comparison covers that omitted phase only at its old noise scale.

1 Conditioning and normalization

Write $v_j = \mathbb{E}U_j^2$, and let W_j be the sum of j independent U_1 variables and $n-j$ independent U_0 variables. Put $\pi_j = \binom{n}{j} p^j q^{n-j}$, $m = \sqrt{n}\pi_k$, and

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k},$$

$$O = \mathbb{P}(W_k > u) + f(u), \qquad T = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u - i).$$

For any upper bound R on π_{k+i}/π_k, conditioning gives

$$\Delta_C \le -m(O - RT) + \mathcal{R}, \quad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho). \tag{1}$$

Indeed the own count loses $\pi_k \mathbb{P}(W_k > u)$, each larger count can add only its closed incoming probability, and all other outgoing terms are nonpositive. This accounts for atoms at every threshold.

Set $A = 3/5$, $v_* = 6/25$, $N = 42$, and $\Xi = (46/25)(251/500)$. We have $\sqrt{v+s} < 251/500$, $|\delta| < \Xi + |u| \le 413/250$, $1 \le k \le n-1$, and $9/25 < k/n < 23/40$. Finite Stirling expansion gives, for $|\delta| \le D$,

$$m = \frac{\phi(0)}{\sqrt{v}} \exp\{-\delta^2/(2nv) + e\}, \qquad n|e| < Z(D) := 5D/8 + 3/8 + D^3/42. \tag{2}$$

For completeness, on the segment $[9/25, 23/40]$ the third derivative of the binary relative entropy has absolute value less than 6, the derivative of $\log(x(1-x))$ has absolute value less than $5/4$, and the combined factorial remainder has absolute value less than $3/(8n)$. Taylor expansion at p gives the three displayed error terms. The last assertion follows from $0 < \log(m!) - (m+1/2)\log m + m - \frac{1}{2}\log(2\pi) < 1/(12m)$ and $x(1-x) > .23$; the two subtracted remainders dominate the remainder at n, or their sum alone bounds the absolute difference. These statements use only $k, n-k > 0$. The derivative constants are checked from $x(1-x) > .23$ and $|1-2x| \le .28$.

For the same fixed δ,

$$\log f'(t) = -\tfrac{1}{2}\log(1 + s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+t)^2}{2n(v+s)} - e.$$

It follows, for $|t| \le b$, that

$$n|\log f'(t)| \le \tfrac{8}{3}\lambda + (2Db + b^2)/(2v_*) + Z(D). \tag{3}$$

For a positive phase, the better signed estimate $2\delta t + t^2 \le \max(0, 2\Xi u - u^2)$, $0 \le t \le u$, may replace $2Db + b^2$ when proving a lower bound for f'. To see this, write $\delta = \xi - u$, $|\xi| \le \Xi$; the convex quadratic in t is bounded by its two endpoint values. The resulting bound increases for $0 \le u \le 91/125 < \Xi$.

Using $.39886 < \phi(0) < .4$, $e^{-x} \ge 1 - x$ and $e^x \le (1-x)^{-1}$ gives the following rational estimates. Every entry is a strict bound except the variance column. The value of f' is needed only between 0 and the actual u.

phase	D	m lower	f'	R upper	$\mathcal{R}$ upper
$[0, .075]$	1	.73	$> .96$	1.06	$(.75)\lambda/n$
$[-.272, -.15]$	1.2	.71	< 1.08	1.08	$(1.1)\lambda/n$
$[.075, .15]$	1.075	.73	$> .95$	1.07	$(.88125)\lambda/n$
$[.15, .728]$	1.652	.65	$> .9$	1.143	$(1.891)\lambda/n$

Here the binomial ratio at step j equals $1 + (p - \delta - j)/(nv + q(\delta + j))$. For the first three rows only the first factor can exceed one; bound it by $1 + (.53 + D - 1)/(42v_* - .6(D-1))$. For the last row multiply this bound by $1 + (.53 + D - 2)/(42v_*)$; all later factors are at most one.

We justify the repair column. Conditional signs are fixed since $a_0 < p$ and $a_1 < q$. Cubic expansion and $U_0^3 \le a_0 U_0^2$, $U_1^3 \ge -a_1 U_1^2$ give

$$\mathbb{E}|Y - p|^3 \ge v(1-2v) + (3\min(p,q) - a_0 - a_1)s.$$

The elementary bound $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$ therefore yields

$$(v+s)^{3/2}(\rho - \rho_B) \ge s(87/250 - 13s/16).$$

Since $(2/5)(87/250 - 13U_*/(16N)) > (251/500)^3$, we have $C(\rho - \rho_B) > s$. Differentiating the normal term in its variance shows it is at most $(7/4)\delta_+ s$. Thus

$$\mathcal{R} < (7D/4 - 1)\lambda/n. \tag{4}$$

No higher moment of an unconfined law has been used.

2 A factorial bound for the complete incoming tail

We use the quadratic-hinge Poisson comparison of Pinelis, *On the Bennett–Hoeffding inequality*, arXiv:0902.4058, Proposition 2.8 and Corollary 2.11. For independent centered summands bounded above by A and with total variance at most V, it gives

$$\mathbb{P}(Z \geq b) \leq \frac{\mathbb{E}(P_\mu - j)_+^2}{(b/A + \mu - j)^2}, \quad \mu = V/A^2, \quad b/A + \mu - j > 0,$$

where P_μ is Poisson. Variance inflation is allowed by the theorem; we never assume that the displayed quotient is monotone in μ.

As $|v_1 - v_0| \leq (5/2)s$, the count at distance i has variance at most $\lambda[1 + (5/2)(D+i)/n]$. More importantly, every legal count has variance at most

$$n \max(v_0, v_1) \leq (5/2)\lambda. \tag{5}$$

Use the first bound for one or two initial counts, then the second bound for all the rest.

Lemma 2. *With $\mu = (5/2)\lambda/A^2$, let $m_0 \geq 2$ be an integer. If $D_{m_0} := m_0(1/A - 1) - u/A + \mu > 0$ and $\mu < m_0 + 2$, then*

$$\sum_{i=m_0}^{n-k} \mathbb{P}(W_{k+i} \leq u - i) \leq J_{m_0}(\mu, u) := \frac{\mu^{m_0+1}}{(m_0+1)! D_{m_0}^2 (1 - \mu/(m_0+2))}.$$

Proof. Apply the comparison to $-W_{k+i}$, using the common variance bound and the integer hinge $j = i$. For every integer $l \geq 0$ and $i \geq 2$, $(l-i)_+^2 \leq \binom{l}{i+1}$. Indeed, writing $l = i + h$, $h \geq 1$, the right side is at least $\binom{h+2}{3} \geq h^2$; the last inequality is equivalent to $(h-1)(h-2) \geq 0$. Poisson factorial moments then bound the numerator by $\mu^{i+1}/(i+1)!$. The hinge denominators increase with i, so successive terms have ratio at most $\mu/(i+2) \leq \mu/(m_0+2)$. Summing the infinite geometric majorant proves the claim and includes all finite incoming counts. Starting this argument at $i = 1$ is false: at $l = 3$, $4 > \binom{3}{2}$. □

For the first count we use

$$Q(\mu, u) = \frac{\mu^2(1/2 + \mu/6)}{((1-u)/A - 1 + \mu)^2}.$$

This follows from $\mathbb{E}(P_\mu - 1)_+^2 = \mu^2 - \mu + 1 - e^{-\mu} \leq \mu^2/2 + \mu^3/6$, provided its unsquared denominator is positive. For the second count, a variance bound $A^2\mu_2$ gives $\mu_2^3/[6((2-u)/A - 2 + \mu_2)^2]$. All these denominators, and all geometric denominators below, are checked positive on their entire stated domains.

3 The three outer intervals reduce to endpoint inequalities

In this section every variance is inflated to its value at U_*. Put $\mu_G = (5/2)U_*/A^2 = 25/48$. For the negative interval, Cantelli and the normal derivative bound give

$$O \geq L_- := \min_{r \in \{3/20, 34/125\}} \left\{ \frac{r^2}{(15/14)U_* + r^2} - \frac{27r}{25} \right\}.$$

The derivative is positive at $r = 3/20$ and has only a single maximum before decreasing, so a minimum on the interval occurs at an endpoint. The complete tail is at most $Q((8/7)U_*/A^2, -3/20) + J_2(\mu_G, -3/20)$. This uses monotonicity in u only, after fixing the inflated variances.

For the next positive interval $[3/40, 3/20]$, use

$$O - RT \geq \frac{19u}{20} - \frac{107}{100}\left\{Q((9/8)U_*/A^2, u) + J_2(\mu_G, u)\right\}.$$

The right side is concave in u, since its subtracted terms are positive multiples of reciprocals of squared positive affine functions. Its two endpoints suffice.

For $3/20 \leq u \leq 91/125$, set $D = 413/250$, $t_1 = (7/6)U_*$ and $\mu_2 = U_*[1 + (5/2)(D + 2)/42]/A^2$. Cantelli bounds the first incoming event, giving

$$O - RT \geq \frac{9u}{10} - \frac{1143}{1000}\left\{\frac{t_1}{t_1 + (1-u)^2} + \frac{\mu_2^3}{6((2-u)/A - 2 + \mu_2)^2} + J_3(\mu_G, u)\right\}.$$

This is again concave: the first tail term has nonnegative second derivative because $3(1 - 91/125)^2 > t_1$, and the others are convex reciprocal squares. Thus two more endpoints suffice. Multiplying by the relevant lower bound for m and subtracting $(7D/4-1)U_*/42+U_*/100$ gives a strictly positive rational number in all five tests. Their minimum exceeds $17/10000$. Each test also proves $O - RT > 0$, as needed when replacing m by its lower bound. The use of U_* in the repair and desired margin is conservative for every $0 < \lambda \leq U_*$.

4 The positive inner interval is one polynomial calculation

Here $0 \leq u \leq 3/40$. Put $\alpha = 16/25$, $M = 5\alpha^4$ and $t = \mathbb{E}W_k^2$. The quartic

$$h(x) = (-x^4 + 3\alpha^2 x^2 + 2\alpha^3 x)/M \leq \mathbf{1}_{(0,\infty)}(x)$$

has numerator $-x(x+\alpha)^2(x-2\alpha)$ and positive-half-line maximum $(9+6\sqrt{3})/20 < 1$. Since $\mathbb{E}W_k^3$ has absolute value at most At and $\mathbb{E}W_k^4 \leq A^2 t + 3t^2$, expansion at $W_k - u$ and $f(u) \geq (24/25)u$ yield

$$O \geq L(t,u) := \frac{(3\alpha^2 - A^2)t - 3t^2 - 4Aut - 6u^2 t - u^4 + 3\alpha^2 u^2 - 2\alpha^3 u}{M} + \frac{24u}{25}. \tag{6}$$

This is concave in t, and $(79/84)\lambda \leq t \leq (89/84)\lambda$. Use the actual λ in the tail, with $\mu_1 = 25\lambda/8$ and $\mu_G = 125\lambda/18$. It remains to show, for each $c \in \{79/84, 89/84\}$,

$$\frac{73}{100}\{L(c\lambda, u) - \tfrac{53}{50}[Q(\mu_1, u) + J_2(\mu_G, u)]\} - (\tfrac{3}{168} + \tfrac{1}{100})\lambda > 0. \tag{7}$$

Here the $3/168$ is the normal repair after the unit cubic gain.

This last verification is a small exact polynomial certificate. Set

$$d_1 = 2/3 - 5u/3 + 25\lambda/8, \quad d_2 = 4/3 - 5u/3 + 125\lambda/18, \quad d_3 = 1 - 125\lambda/72.$$

Multiply the left side of (7) by $6d_1^2 d_2^2 d_3 > 0$. The result $P_c(\lambda, u)$ has degree at most 7 in λ and 8 in u. Write $P_c = \sum a_{kl}\lambda^k u^l$. Its tensor Bernstein coefficients on the single square $[0, 3/40]^2$ are exactly

$$b_{ij} = \sum_{k \leq i, l \leq j} a_{kl}(3/40)^{k+l} \frac{\binom{i}{k}}{\binom{7}{k}} \frac{\binom{j}{l}}{\binom{8}{l}}, \qquad 0 \leq i \leq 7, \quad 0 \leq j \leq 8.$$

The source `outer-check.py` constructs these polynomials from (6)–(7) using rational arithmetic, and checks all 144 coefficients: $b_{00} = 0$ and every other coefficient is strictly positive, for both values of c. The transform follows from $x^k = \sum_{i=k}^m \binom{i}{k}/\binom{m}{k} B_i^m(x)$. Bernstein basis functions are nonnegative and sum to one. At every point of the square except $(0,0)$ at least one nonzero-index basis function is positive. Consequently $P_c > 0$ whenever $\lambda > 0$, including all boundary phases. This proves (7) and completes the theorem.

5 What the check does and does not establish

The acceptance script checks the scalar normalization and endpoint inequalities and reconstructs both coefficient arrays. Missing, repeated, reordered and corrupted arrays are rejected; removing the cubic gain or increasing the noise endpoint to .1 fails a genuine sufficient inequality. The proof above supplies the analytic reduction; an arithmetic PASS alone supplies neither that reduction nor acquisition of its premises. The independent review and saved runs are separate files. No inherited certificate or manuscript is replaced by this component.